

Lesson Notes

MODULE 5 | LESSON 3

SAMPLING AND INFERENCE

Reading Time	25 minutes
Prior Knowledge	Basic probability and statistics
Keywords	Inference, Belief propagation, Direct Sampling, Variable elimination, Gibbs sampling, Weighted sampling

In the last lesson, we got more familiar with Bayesian networks by working through two sets of code that implement the basic idea, wherein both sets of code tackled the same problem: the Monty Hall Problem. In this lesson, we go back to the algorithms textbook to understand the techniques of inference and sampling so that we might answer the following question: What am I supposed to do with all this data?

1. Bayesian Inference

This lesson's reading runs the gambit. It introduces some new foundational concepts in a thoughtful way; it glosses certain concepts with which you should already be familiar from previous coursework; it mentions some techniques that we will revisit in the next lesson and examine in more detail there; but at least twice, it breezes through some more technical or otherwise obscure ideas in a way that some students might find mystifying. In the following few pages, we attempt to provide some helpful clarification around Bayesian inference as well as enough additional information about belief propagation to at least whet your appetite to learn more.

As in many of the tasks we have set ourselves in this practitioner track, we need to define a model based on the available data that best predicts or classifies. This is clear in machine learning and perhaps even more transparent in deep learning. On this topic, we can say with Xing that learning "involves estimating parameters of the model M so that it explains or 'fits' the data D " (1). Moreover, when our data is not complete, we must use inference to find—i.e., infer—the hidden data.

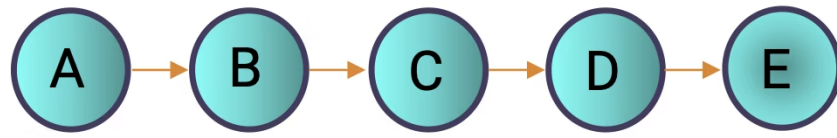
When working with Bayes' rule, which we do when we build Bayesian networks, this means working with the following four probabilities, but in a way that is specific to the inference task at hand:

1. Assigning prior probability distributions over the parameters $P(\theta)$, representing the probability of the hypothesis θ being true before applying the Bayes' theorem. This assignment is not in fact *Bayesian* in the sense of the word that implies not frequentist: The prior probability can certainly be informed by frequentist statistical methods (Jihan).
2. Calculating the likelihood $P(X|\theta)$, which equals the conditional probability of the evidence given a hypothetical parameter set, θ . In the context of inference, it is the marginal probability of a subset of the variables conditioned on the evidence (Xing 1).
3. The evidence term $P(X)$ equals the probability of evidence or data: "This can be expressed as a summation (or integral) of the probabilities of all possible hypotheses weighted by their likelihood" (Jihan).
4. $P(\theta|X)$ - Posteriori probability denotes the conditional probability of the hypothesis θ after observing the evidence X (Jihan). In probabilistic inference, it refers to the "posteriori belief," which is "the conditional probability distribution of some query nodes conditioned on the evidence"; when we maximize this value, we have found the "most probable assignment" (MPA) (Xing 4, 1).

2. Variable Elimination

Actually, we have oversimplified things slightly with regard to the MPA in the fourth point. In fact, according to Xing, "a key observation is that given query and evidence nodes" in our Bayesian network, calculating the posteriori belief does not necessarily involve the entire set of evidence (1–2). Summing out variables we consider irrelevant is called marginalization, and relieves us of having to try to calculate the higher-dimensional distribution that would otherwise include them in the calculation (Xing 4, 1–2).

Figure 1: A Simple Directed Probabilistic Model with Chain Structure



For example, see Figure 1 and assume that each of the five random variables represented there can take n number of values, which could require “a full joint probability table containing n^5 entries” to fully describe (3) using the formula

$$P(e) = \sum_{a,b,c,d} P(a,b,c,d,e)$$

This would require summing out, or marginalizing, four variables. We observe, however, that we can use the chain rule to factorize this joint probability, as follows:

$$\begin{aligned} P(e) &= \sum_{a,b,c,d} P(a,b,c,d,e) \\ &= \sum_{a,b,c,d} P(a)P(b|a)P(c|b)P(d|c)P(e|d) \end{aligned}$$

Kochenderfer refers to this as the “factor product” operation. We observe that “not all terms have a dependence on all variables” (Kochenderfer) in this factorized joint distribution. This means that we can sum over A without affecting three of the five distributions, which in turn means that an equivalent formula is the product of these two distinct sums, one sum over the product of two terms involving a , and one sum over a product of terms not involving a :

$$\begin{aligned} P(e) &= \sum_{a,b,c,d} P(a)P(b|a)P(c|b)P(d|c)P(e|d) \\ &= \sum_{b,c,d} P(c|b)P(d|c)P(e|d) \sum_a P(a)P(b|a) \\ &= \sum_{b,c,d} P(c|b)P(d|c)P(e|d)P(b) \end{aligned}$$

As you can see from the last line, we have eliminated a variable: We now have only four terms (none with a) and one summation—that is, until we repeat this process, this time marginalizing the two terms that involve b and leaving the other two terms intact (for now).

$$\begin{aligned} P(e) &= \sum_{b,c,d} P(c|b)P(d|c)P(e|d)P(b) \\ &= \sum_{c,d} P(d|c)P(e|d) \sum_b P(c|b)P(b) \\ &= \sum_{c,d} P(d|c)P(e|d)P(c) \end{aligned}$$

By the time we get to the last line, as you can see, we are left with only three terms. Obviously, when we repeat the variable elimination (VE) twice more, we have arrived at the probability result we were seeking. You will see another, only slightly more complicated example of the factor product followed by factor marginalization in Kochenderfer (45–46) and the relevant algorithms for each operation. Hopefully, this simple example prepares you for those. Kochenderfer also defines factor conditioning (44); this really is as straightforward as the name implies, though that definition may be confusing. It means that we are inferring a conditional probability. A conditional probability is just a probability conditioned on some temporary assumption, which is what Kochenderfer confusingly (though correctly) refers to as “inconsistent with the evidence” (44).

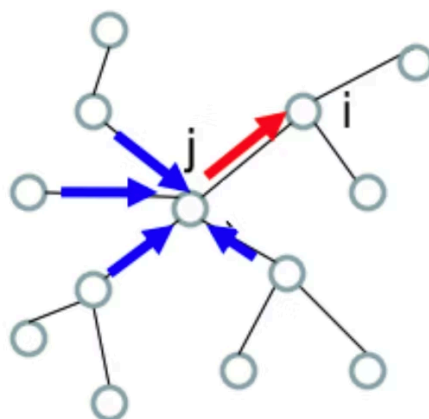
In the last lesson, we reminded ourselves about factors and their relationship to random variables in factor graphs: Factors are real-valued functions of a subset of the random variables with which we are concerned. We also saw how the normalized product of the factors gives us the posterior probability distribution of the random variables. You should assume that the probabilities in the above series of marginalizations are precisely these factors.

3. Belief Propagation

Belief propagation is the same as VE except that it includes passing intermediate tables called “messages” in each of the elimination steps. What are these “messages”? The messages are just the solutions to “a set of self-consistent equations satisfied by the variables or probabilities of interest and then solving those equations by numerical iteration (Kirkley 1), and they provide the sought marginal distributions (Stratos 4). With each individual and independently computed query, “several intermediate terms may be computed repeatedly” and the message passing allows for the sharing of the intermediate terms, and this reuse of information makes the algorithm much more efficient (Xing 13, 1). Notice for example in figure 2 how the

messages carried by the blue arrows are the same for many distinct queries depending on the conditions; their calculation might therefore be redundant, so we can save computational resources by skipping when it is reasonable, which is what belief propagation does.

Figure 2: Belief Propagation Example



Source: Caravelli, Francesco, and Cristiano Nisoli. "Computation via Interacting Magnetic Memory Bites: Integration of Boolean Gates." *arXiv*. https://www.researchgate.net/publication/328446390_Computation_via_Interacting_Magnetic_Memory_Bites_Integration_of_Boolean_Gates

4. Conclusion

In this lesson, we examined several techniques for inference and sampling, preparing us for the next lesson where we put these techniques to work in Python.

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